Blueprint: Lattice Yang–Mills Mass Gap at Strong Coupling

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Abstract

We describe a formal proof in Lean 4 of exponential correlation decay (mass gap) for Wilson's lattice gauge theory with compact gauge group $G \subseteq U(n)$ on the periodic lattice $(\mathbb{Z}/N\mathbb{Z})^d$, $d \geq 2$, at strong coupling. The proof follows the Dobrushin uniqueness approach of Chatterjee (2026), Chapter 16, and comprises ∼12,000 lines of new Lean 4 code with zero `sorry`s and zero custom axioms in the main `lgt` repository.

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1 Overview

The proof proceeds in three stages:

Stage A (Parts 1–4): Encode lattice gauge theory as a Gibbs specification on the link lattice, and prove the Yang–Mills measure satisfies the DLR equations.

Stage B (Parts 5–7): Verify Dobrushin's condition at strong coupling and construct the Dobrushin coupling, yielding exponential decay of the disagreement vector via the Neumann series.

Stage C (Parts 8–9): Transfer the coupling-level decay to a covariance bound on plaquette observables via multi-site disintegration.

2 Part 1: The lattice as a cell complex

Lean: `LGT/Lattice/CellComplex.lean`

We work on the d-dimensional periodic lattice $(\mathbb{Z}/N\mathbb{Z})^d$, equipped with an explicit CW-structure:

- **Sites** (0-cells): points $x = (x_1, \ldots, x_d) \in (\mathbb{Z}/N\mathbb{Z})^d$.
- **Links** (1-cells): oriented edges $\ell = (x, \mu)$ where x is a site and $\mu \in \{0, \ldots, d-1\}$ is a lattice direction. The link connects x to $x + e_\mu \pmod N$.
- **Plaquettes** (2-cells): oriented unit squares $p = (x, \mu, \nu)$ with $\mu < \nu$. The boundary ∂p consists of four links traversed counterclockwise:

$$\partial p = \{(x, \mu),\ (x{+}e_\mu, \nu),\ (x{+}e_\nu, \mu),\ (x, \nu)\}.$$

  The third and fourth links are traversed with reversed orientation.

Two links share a plaquette iff they are edges of a common unit square. On the standard lattice, each link lies on at most $2(d-1)$ plaquettes, and two distinct links share at most one plaquette.

3 Part 2: Gauge fields and the Wilson action

Lean: `LGT/GaugeField/`, `LGT/WilsonAction/`

Fix a compact matrix Lie group $G \subseteq U(n)$ with the fundamental representation $\rho\colon G \to M_n(\mathbb{C})$.

A **gauge configuration** (or connection) is a function $U\colon \text{Links} \to G$ assigning a group element to each link. The configuration space is $G^{|\text{Links}|}$, a compact product of copies of G.

The **holonomy** around a plaquette p is the ordered product of group elements around its boundary (cf. Chatterjee §15.2):

$$U_p = U(\ell_1) \cdot U(\ell_2) \cdot U(\ell_3)^{-1} \cdot U(\ell_4)^{-1}.$$

For a flat connection ($U_p = 1$), the holonomy is trivial. The holonomy measures the discrete curvature.

The **Wilson plaquette action** (Wilson 1974) is:

$$S(U) = \beta \sum_{p \in \text{plaq}} (n - \operatorname{Re}\operatorname{Tr}(U_p))$$

where $\beta \geq 0$ is the inverse coupling constant.

Lemma 1 (`unitaryGroup_gaugeReTr_le`)**.** *For $U \in U(n)$, $|\operatorname{Re}\operatorname{Tr}(U)| \leq n$.*

Proof. Each diagonal entry satisfies $|U_{ii}|^2 \leq \sum_k |U_{ki}|^2 = 1$ (column norm one), hence $|\operatorname{Re} U_{ii}| \leq |U_{ii}| \leq 1$, and $|\operatorname{Re}\operatorname{Tr}(U)| = |\sum_i \operatorname{Re} U_{ii}| \leq n$. □

The plaquette cost $n - \operatorname{Re}\operatorname{Tr}(U_p) \in [0, 2n]$, so $S \geq 0$ and the **Boltzmann weight** $w(U) = e^{-S(U)} \in (0, 1]$.

The action is **gauge invariant**: under a gauge transformation $g\colon \text{Sites} \to G$ acting by $U(\ell) \mapsto g(\text{source}) \cdot U(\ell) \cdot g(\text{target})^{-1}$, the holonomy transforms as $U_p \mapsto g(x) \cdot U_p \cdot g(x)^{-1}$, and $\operatorname{Re}\operatorname{Tr}$ is invariant by cyclicity of trace (cf. Chatterjee §15.3).

4 Part 3: The Yang–Mills measure

Lean: `LGT/MassGap/YMMeasure.lean`

The **prior measure** on configurations is the product Haar probability measure:

$$\mu_{\text{Haar}} = \bigotimes_{\ell \in \text{Links}} \text{Haar}_G.$$

The **Yang–Mills measure** is the Boltzmann-weighted measure (cf. Chatterjee §15.4):

$$\mu_{\text{YM}}(dU) = \frac{1}{Z}\, e^{-S(U)}\, d\mu_{\text{Haar}}(U)$$

where $Z = \int e^{-S}\, d\mu_{\text{Haar}}$ is the **partition function**.

Lemma 2 (`partitionFn_pos`)**.** *$Z > 0$.*

Proof. $w(U) > 0$ for all U, with $w(U) \geq e^{-\beta \cdot |\text{plaq}| \cdot 2n} > 0$. Since μ_{Haar} is a probability measure, $Z \geq e^{-\beta \cdot |\text{plaq}| \cdot 2n} > 0$. □

The **connected 2-point function** (covariance) of observables f, g is:

$$\langle f \cdot g \rangle - \langle f \rangle \langle g \rangle$$

where $\langle \cdot \rangle$ denotes expectation under μ_{YM}. The mass gap is the statement that this decays exponentially when $f = \operatorname{Re}\operatorname{Tr}(U_p)$ and $g = \operatorname{Re}\operatorname{Tr}(U_q)$.

5 Part 4: The DLR identity — Yang–Mills as a Gibbs measure

Lean: `LGT/Gibbs/YMSpec.lean`, `LGT/Gibbs/YMIsGibbs.lean`

The strategy for proving correlation decay is to apply the Dobrushin uniqueness theorem from the theory of Gibbs measures (Chatterjee §16.3, Georgii §8).

5.1 Gibbs specifications

A **Gibbs specification** (Georgii §1.2, Chatterjee §16.1) is a rule that, given any finite region Λ and any configuration σ on Λ^c, produces a conditional probability distribution on Λ-configurations.

For lattice gauge theory, the "sites" are links and the "spin space" is the gauge group G. Given a finite subset $\Lambda \subseteq \text{Links}$ and a boundary condition $\sigma\colon \text{Links} \to G$, the Gibbs conditional is:

$$\gamma(\Lambda, \sigma)(A) = \frac{1}{Z_\Lambda(\sigma)} \int_{G^\Lambda} \mathbf{1}_A\big(\text{glue}(\Lambda, u_\Lambda, \sigma)\big)\, e^{-S(\text{glue}(\Lambda, u_\Lambda, \sigma))}\, d\,\text{Haar}^\Lambda(u_\Lambda).$$

The **gluing operation** $\text{glue}(\Lambda, u_\Lambda, \sigma)$ produces a full configuration by combining fresh integration variables u_Λ on the links in Λ with the fixed boundary values σ on the links outside Λ:

$$\text{glue}(\Lambda, u_\Lambda, \sigma)(\ell) = \begin{cases} u_\Lambda(\ell) & \text{if } \ell \in \Lambda, \\ \sigma(\ell) & \text{if } \ell \notin \Lambda. \end{cases}$$

The integral is over the product Haar measure Haar^Λ on G^Λ, and $Z_\Lambda(\sigma) = \int w(\mathrm{glue}(\Lambda, u_\Lambda, \sigma))\, d\,\mathrm{Haar}^\Lambda$ is the **conditional partition function**, normalizing the density to a probability measure.

The specification satisfies three structural axioms:

- **Consistency**: $\gamma(\Lambda, \sigma)$ depends on σ only through $\sigma|_{\Lambda^c}$.
- **Properness**: $\gamma(\Lambda, \sigma)$ is concentrated on configurations agreeing with σ outside Λ.
- **Measurability**: $\sigma \mapsto \gamma(\Lambda, \sigma)(A)$ is measurable for each A.

5.2 The DLR equation

A probability measure μ is a **Gibbs measure** for γ if it satisfies the **DLR equation** (Dobrushin–Lanford–Ruelle, Georgii §1.5): for every finite Λ and measurable A,

$$\mu(A) = \int \gamma(\Lambda, \sigma)(A)\, d\mu(\sigma). \tag{1}$$

In words: the probability μ assigns to any event A equals the average, over boundary conditions σ drawn from μ itself, of the conditional probability $\gamma(\Lambda, \sigma)(A)$. On a finite lattice with Dobrushin's condition, (1) has a unique solution; on infinite lattices, multiple solutions may exist (phase coexistence).

5.3 Proving the DLR identity

We prove that μ_{YM} satisfies (1) — this is `ymMeasure_isGibbs`, the most technically demanding theorem in the formalization. The proof assembles five intermediate results:

1. **Pushforward identity** (`glue_measurePreserving`). The map $(u, \sigma) \mapsto \mathrm{glue}(\Lambda, u, \sigma)$ pushes the product measure $\mu_{\mathrm{Haar}} \times \mu_{\mathrm{Haar}}$ forward to μ_{Haar}. Proved via `Measure.pi_eq` on measurable boxes.
2. **Fubini helper** (`integral_glue_split_eq`). For integrable F:
$$\int_\sigma \int_{u_\Lambda} F(\mathrm{glue}(u, \sigma))\, d\,\mathrm{Haar}(u)\, d\,\mathrm{Haar}(\sigma) = \int_U F(U)\, d\,\mathrm{Haar}(U).$$
3. **Identity A** (`integral_smul_condZ_eq_integral_smul_w`). For h constant on Λ^c-fibers: $\int h \cdot Z_\Lambda\, d\,\mathrm{Haar} = \int h \cdot w\, d\,\mathrm{Haar}$.
4. **Cancellation identity** (`cancellation_identity`). For h fiber-constant: $\int h \cdot w / Z_\Lambda\, d\,\mathrm{Haar} = \int h\, d\,\mathrm{Haar}$.
5. **DLR assembly** (`ymMeasure_dlr`). Combines (4) with the indicator-weighted Fubini identity and the `withDensity` unfolding of μ_{YM} and $\gamma(\Lambda, \sigma)$.

6 Part 5: The Dobrushin condition at strong coupling

Lean: `LGT/Gibbs/YMDobrushin.lean`, `LGT/MassGap/DobrushinVerification.lean`

Dobrushin's condition (Dobrushin 1968, Chatterjee §16.2) provides a criterion for exponential correlation decay.

Definition 3. The **influence coefficient** $C(x, y)$ is:

$$C(x, y) = \sup_{\sigma \sim_y \tau} d_{\mathrm{TV}}\big(\gamma(\{x\}, \sigma)|_x,\ \gamma(\{x\}, \tau)|_x\big)$$

where the supremum is over boundary conditions σ, τ differing only at link y, and $\cdot|_x$ denotes the x-marginal.

Definition 4. Dobrushin's condition holds if $\alpha := \max_x \sum_y C(x,y) < 1$.

For the YM specification, we prove two influence bounds:

- **Off-plaquette**: $C(x,y) = 0$ when x and y don't share a plaquette. The conditional density at x involves only plaquettes through x; the action decomposes as $S = S_x + S_{\text{rest}}$ where S_x doesn't involve y. The density ratio $w(\text{glue}(u,\sigma))/w(\text{glue}(u,\tau))$ is exactly 1.
- **On-plaquette**: $C(x,y) \le 1 - e^{-4n\beta}$ when x, y share a plaquette. The shared plaquette contributes at most $\beta \cdot 2n$ to the energy oscillation, giving an unnormalized density ratio of $e^{\pm 2n\beta}$. After normalization, the measure ratio is $e^{\pm 4n\beta}$, giving TV $\le 1 - e^{-4n\beta}$.

The column sum is at most $\text{maxNeighbors}(d) \cdot (1 - e^{-4n\beta})$, where $\text{maxNeighbors}(d) = 8(d-1)$. For

$$\beta < \frac{1}{4n \cdot 8(d-1)} = \frac{1}{32\,n\,(d-1)},$$

Dobrushin's condition holds with $\alpha < 1$ (cf. Chatterjee Theorem 16.3.1).

7 Part 6: The Dobrushin coupling

Lean: `markov-semigroups/Coupling/DobrushinCoupling.lean`, `markov-semigroups/Coupling/Canonical`

The core of the Dobrushin uniqueness theorem is the construction of a **joint coupling** P of two measures μ_1, μ_2 (both satisfying DLR on a set T) such that the per-site disagreement satisfies the **contraction inequality** simultaneously at all $z \in T$:

$$P\{\sigma_z \neq \eta_z\} \le \sum_w C(z,w) \cdot P\{\sigma_w \neq \eta_w\}.$$

Theorem 5 (`dobrushin_iterated_coupling_fintype`)**.** *Among all couplings of μ_1 and μ_2, let P^* minimize the total disagreement $D(P) = \sum_z P\{\sigma_z \neq \eta_z\}$. Then P^* satisfies the per-site contraction at every $z \in T$.*

Proof. The minimum exists by **Prokhorov compactness**: the space of probability measures on a compact space is compact in the weak topology, and the coupling set is closed. Total disagreement D is lower semicontinuous (each $\{\sigma_z \neq \eta_z\}$ is open for T_2 spaces).

At the minimizer P^*, if $P^*\{\sigma_z \neq \eta_z\} > \sum_w C(z,w)\, P^*\{\sigma_w \neq \eta_w\}$ for some $z \in T$, then replacing the z-coupling with the **maximal coupling** of the z-conditionals produces P' with $D(P') < D(P^*)$ (since the z-disagreement decreases while all other sites' disagreement is unchanged), contradicting minimality. □

The **canonical maximal coupling** of μ, ν on a countable space is:

$$P_{\max} = (\mu \wedge \nu) \cdot \Delta + c^{-1} \cdot (\mu - \mu \wedge \nu) \otimes (\nu - \mu \wedge \nu)$$

where $(\mu \wedge \nu)(\{a\}) = \min(\mu\{a\}, \nu\{a\})$, Δ is the diagonal embedding, and $c = \text{TV}(\mu,\nu)$. This achieves $P\{a \neq b\} = \text{TV}(\mu,\nu)$ and varies measurably in (μ,ν) — crucial for the `Measure.bind` kernel.

8 Part 7: Exponential decay via the Neumann series

Lean: `markov-semigroups/Dobrushin/NeumannSeries.lean`, `markov-semigroups/Dobrushin/CondTVBridg`

Given the coupling with per-site contraction, the disagreement vector $\delta(z) = P\{\sigma_z \neq \eta_z\}$ satisfies:

$$\delta(z) \le \sum_w C(z,w) \cdot \delta(w).$$

This is a matrix inequality $\delta \le C\delta$. Since $\alpha < 1$, iterating gives $\delta \le C^N \delta_0 + \sum_{k<N} C^k \cdot s$ where s is a "source" term from sites outside T. In the limit $N \to \infty$:

$$\delta(y) \le \sum_{x \in N_f} \left((I - C)^{-1}\right)_{yx}$$

where $(I - C)^{-1} = \sum_{n \ge 0} C^n$ is the Neumann series.

For nearest-neighbor influence ($C(z, w) = 0$ when they share no plaquette), the resolvent entry is bounded by (cf. Chatterjee §16.2):

$$\left((I - C)^{-1}\right)_{yx} \le \frac{\alpha^{\mathrm{dist}(y,x)}}{1 - \alpha}$$

giving exponential decay.

9 Part 8: From coupling to covariance

Lean: `markov-semigroups/Dobrushin/CovarianceBoundMultisite.lean`, `markov-semigroups/Dobrushin/`

The plaquette observable $\mathrm{Re}\,\mathrm{Tr}(U_p)$ depends on $|N_f| = 4$ links, not a single link. This requires a **multi-site covariance bound**.

Theorem 6 (`covariance_bound_gibbs_multisite_general_nocount`)**.** *For f depending on $N_f \subseteq$ Links and g depending on N_g, with $|f| \le B_f$ and $|g| \le B_g$:*

$$|\,\mathrm{Cov}_\mu(f, g)| \le 2\, B_f\, B_g \sum_{x \in N_f} \sum_{y \in N_g} \frac{\alpha^{\mathrm{dist}(y,x)}}{1 - \alpha}.$$

The proof has three ingredients:

Ingredient 1: Covariance decomposition via disintegration. Disintegrate μ at the N_f-fiber using Mathlib's `Measure.condKernel` (regular conditional probability for Standard Borel spaces). Since f is N_f-local:

$$\mathrm{Cov}(f, g) = \int f(b) \cdot \left(E[g \mid \mathrm{fiber} = b] - E[g]\right) d\mu_{\mathrm{fst}}(b).$$

Bounding $|f| \le B_f$ gives $|\,\mathrm{Cov}\,| \le B_f \cdot \sup_b |E[g|b] - E[g]|$.

For uncountable compact G (like $U(n)$), the discrete fiber decomposition fails (fibers have measure zero). We use Mathlib's `condKernel` disintegration, which works for arbitrary Standard Borel spaces via Prokhorov's theorem.

Ingredient 2: Conditional expectation bound via coupling. For each fiber value b, the conditional measure μ_b (the `condKernel` fiber lifted back to the configuration space) satisfies DLR at sites $z \notin N_f$:

Theorem 7 (`fiberMeasure_dlr_ae`)**.** *For a.e. b, μ_b satisfies* (1) *at every $\{z\}$ with $z \notin N_f$.*

Proof. Transport the DLR equation for μ through the disintegration $\rho = \rho_{\mathrm{fst}} \otimes_\kappa \rho_{\mathrm{condKernel}}$. For each measurable A, the set-integral identity $\int_T F_A(b)\, d\rho_{\mathrm{fst}} = \int_T G_A(b)\, d\rho_{\mathrm{fst}}$ follows from DLR on restricted sets and `condDist` properness. Then `Integrable.ae_eq_of_forall_setIntegral_eq` gives $F_A = G_A$ a.e. for each A. The upgrade from "for each A, a.e. b" to "a.e. b, for all A" uses the countable generating π-system from `StandardBorelSpace` (via `MeasurableSpace.ae_induction_on_inter`). Finite intersection over $z \notin N_f$ is trivial (I is finite). □

With DLR for μ_b, the Dobrushin coupling applies, giving:

$$|E[g \mid b] - E[g]| \le 2\, B_g \sum_{y \in N_g} \sum_{x \in N_f} \left((I - C)^{-1}\right)_{yx}.$$

Ingredient 3: Nearest-neighbor exponential decay. Substituting the resolvent bound and noting $\mathrm{dist}(y,x) \geq \mathrm{dist}(p,q) - 2$:

$$|\,\mathrm{Cov}\,| \leq 2\, B_f\, B_g \cdot |N_f| \cdot |N_g| \cdot \frac{\alpha^{\mathrm{dist}(p,q)-2}}{1-\alpha}.$$

10 Part 9: Assembly

Lean: `LGT/MassGap/StrongCoupling.lean`, `LGT/GaugeField/UnitaryGroup.lean`

`ym_mass_gap_strong_coupling` wires Parts 1–8:

1. Build $\gamma =$ `ymGibbsSpec`.
2. Verify the Dobrushin condition: $\alpha < 1$ at strong coupling.
3. Prove μ_{YM} is Gibbs for γ (`ymMeasure_isGibbs`).
4. Prove plaquette observables are 4-link-local.
5. Bound the influence coefficients (off-plaquette $= 0$ via action factorization; on-plaquette $\leq 1 - e^{-4n\beta}$ via density ratio).
6. Derive the `condKernel` a.e. bound from `fiberMeasure_dlr_ae`.
7. Apply the multi-site covariance bound.
8. Convert `connected2pt` $\leftrightarrow$ covariance via `ymExpect_eq_integral_ymMeasure`.

`ym_mass_gap_UN` specializes to $G = U(n)$, supplying:

- Trace bounds $|\,\mathrm{Re}\,\mathrm{Tr}(U)| \leq n$ (column-norm-one).
- Continuity of ρ (`continuous_subtype_val`).
- `CompactSpace` (closed + bounded in $M_n(\mathbb{C})$, Heine–Borel).
- `SecondCountableTopology` (subspace of $M_n(\mathbb{C})$).
- `HasHaarProbability` (Mathlib's `haarMeasure` with $K_0 = U(n)$, normalized via `haarMeasure_self` $= 1$).

11 Relation to Chatterjee's exposition

Chatterjee	This formalization
§16.1 Gibbs specifications	`ymGibbsSpec` (YMSpec.lean)
§16.2 Dobrushin uniqueness	`dobrushin_iterated_coupling_fintype`
§16.2 Influence coefficients	`influenceCoeff`, influence bounds
§16.2 Neumann series	`neumannSeriesCoeff`, resolvent bound
§16.3 Application to YM	`ym_mass_gap_strong_coupling`
Theorem 16.3.1	`ym_mass_gap_UN`

Main differences from Chatterjee:

- Multi-site covariance bound (plaquette observables depend on 4 links, not 1).
- Minimum-disagreement coupling (Prokhorov) instead of iterative sweep.
- Explicit DLR proof for μ_{YM}.
- Uncountable gauge groups via Mathlib's `condKernel`.

12 What remains

- **2 lattice combinatorics hypotheses** (provable, ∼160 lines): `hSharedPlaqBound` (two links share ≤ 1 plaquette) and `hPlaqPerLink` (each link borders $\leq 2(d-1)$ plaquettes).
- **1 upstream axiom** (`dobrushin_coupling_axiom_compact`): the minimum-disagreement coupling exists for compact (not just countable) S. Needs Portmanteau lower semicontinuity (∼300 lines).

13 Statistics

Metric	Count
New Lean 4 code (both repos)	∼12,000 lines
Files created	8 (lgt) + 6 (markov-semigroups)
Sorry count (lgt)	0
Sorry count (markov-semigroups)	0
Custom axioms (lgt)	0
Custom axioms (markov-semigroups)	1 (compact coupling)
Hypotheses in final theorem	6 physics + 2 lattice
Upstream bugs found + fixed	2

References

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5. K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10** (1974), 2445–2459.